

Electroadhesives

Aptly named, Grabit Inc. is a company that has built an electrically-controllable adhesion technology that can be used in biomedical lab automation, medical devices and surgical equipment. What is it? Electroadhesion uses electrostatic forces between substrate materials and electroadhesive surface to induce adhesion between materials through a set of conductive electrodes deposited on the surface of a polymer. Who knew that someone could set up a whole company simply on the basis of 'Opposite charges attract.'

pacemaker called Wireless Cardiac Stimulation System (WiCS). It allows for bi-ventricular pacing where a traditional pacemaker paces the right ventricle and EBR WiCS electrode paces the left ventricle.

Transforming the data collected above into graphical form can be done using embedded graphics solutions like AMD Embedded Radeon HD 7850 GPU. Based on Graphic Next Core (GCN) architecture that accelerates medical imaging, it provides three times improved performance with detailed medical image visualisation and other advanced graphics-driven capabilities.

Detecting substances

CAP-100 capacitive liquid level sensor from Gems Sensors & Controls allows level sensing through a wide variety of materials. These compact sensors can detect substances like waste, reagent, buffer or diluents, which would be useful for vessels containing bio-hazardous liquids in a sealed container. These can also be calibrated for use as proximity sensors.

Some manufacturers are also looking at making their components more rugged, such as FlexiForce Sustained Stability Series 301 from Tekscan. Made from pressure-sensitive ink, these can withstand and measure force and pressure in a wider range of environments. It is claimed that reducing drive voltage or feedback resistance can extend the range further.

Other novel uses include the use of temperature sensors to ensure the best application of treatments in dermatology. One example is where SynerionVelaShape III uses Exergen's non-invasive infrared (IR) temperature sensor placed inside the

applicator to continuously monitor skin temperature, while allowing the treatment to continue unhindered.

Moore's Law keeps driving integration

For more integrated components, Honeywell's Humidicon digital humidity and temperature sensors integrate multiple functions in a single compact package. These can be used in respiratory therapy, medical incubators and medical micro-environment applications.

Engineers at Georgia Tech Research Institute have designed a sensor that can detect chemical vapour using its package of three sensors and a radio frequency identification (RFID) chip. Current design of the chip requires power from an incoming signal beam to enable sending data out, but researchers hope future models would be able to use ambient energy.

Engineers usually have to face difficulties when building systems out of separate pump motors, gearboxes and drives due to design complications and uncertainty that comes with separate components. With the aim to reduce the cost for engineers at original equipment manufacturers, Watson-Marlow Pumps Group has brought out a brushless DC gear motor with a fully-integrated speed controller that can be used in medical devices.

Building systems enabling better healthcare

The Internet of Things (IoT) is driving the creation of Web-connected medical devices that keep track of your health and medication, prompting your doctor about potential threats or medical events. One such

device is Vaica's SimpleMed+ medication compliance device that uses Telit Wireless Solutions' GC864-QUAD V2 for mobile connectivity that allows quad-band connectivity using a mobile network.

In March 2015, Redbend and Telit also announced their partnership that allows designers to avoid IT integration at their end if they use the hosted service to manage their machine-to-machine (M2M) devices.

Also interesting is an electronic patch designed by Sensium that can check a patient's vital signs every two minutes and wirelessly send it to authorised medical devices. The patch itself is a low-power 915MHz wireless unit that can measure heart rate, respiratory rate and auxiliary temperature.

Maxim Integrated launched a reference design for sensing galvanic skin response (GSR) in mobile medical and fitness applications with improved accuracy. Offered in a wristband form factor, MAXREFDES73# includes body surface temperature readings, Bluetooth communications and a rechargeable battery that lasts up to one week on a single charge.

Google Glass could be helpful

Electronic health records blended with wearable electronics show promise to enable doctors to improve the time they spend with their patients. Texas based Pristine's app EyeSight lets doctors transmit live video of wounded patients from Google Glass to computers and phones.

There is also Augmedix, a system that would roughly translate information from Glass's audio-visual stream directly into a patient's medical record. And Healium is developing an app that would let doctors share patient information through Glass.

There is also some buzz about a helmet-borne system developed by Vijay Varadan and his research